

Date: 17.09.2024

Roll No. \_\_\_\_\_

III SEMESTER B.Tech (ECE, EIOT)

MID-SEMESTER EXAMINATION 2024

Course Code : ECECC302/EIECC302  
Time: 1:30 Hrs

Course Title : Signals & Systems  
Max Marks: 20

Note: -For Re-registration ie ECECC05 / EIECC05, marks shall be scaled upto 25 marks. Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks | CO  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| Q 1    | <p>a Let <math>h(t)</math> be a triangular pulse defined as</p> $h(t) = \begin{cases} 2-t, & 0 < t \leq 2 \\ 0, & \text{elsewhere} \end{cases}$ <p>and <math>x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)</math>.</p> <p>Determine <math>y(t) = h(t) * x(t)</math> for <math>T=1</math> and <math>T=2</math>. Plot <math>y(t)</math>. Is <math>y(t)</math> periodic for both values of 'T'? If yes what is the time period in both cases.</p>                                                                                                             | 03    | CO1 |
|        | <p>b Show that if <math>x(t)</math> is periodic with time period T, then</p> $\int_a^b x(t) dt = \int_{a+T}^{b+T} x(t) dt \quad \text{and} \quad \int_0^T x(t) dt = \int_a^{a+T} x(t) dt.$                                                                                                                                                                                                                                                                                                                                                                   | 01    | CO1 |
| Q 2    | <p>a Determine whether the system described by <math>y[n] = \sum_{k=-\infty}^{n-1} kx[k+1]</math> is linear, time invariant and causal.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  | 03    | CO1 |
|        | <p>b Find <math>\int_{-\infty}^{\infty} \delta(at - b) \sin^2(t - 4) dt</math>, where 'a' is a positive constant.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 01    | CO1 |
| Q 3    | <p>For each of the following statements determine whether it is true or false. Justify your answer.</p> <p>i. If <math>x[n]=0</math> for <math>n &lt; N1</math>, and <math>h[n]=0</math> for <math>n &lt; N2</math>, then <math>x[n]*h[n]=0</math> for <math>n &lt; N1+N2</math>.</p> <p>ii. If <math>y(t) = x(t)*h(t)</math>, then <math>y(-t) = x(-t)*h(-t)</math></p> <p>iii. If <math>x(t)=0</math> for <math>t &gt; T1</math>, and <math>h(t)=0</math> for <math>t &gt; T2</math>, then <math>x(t-1)*h(t+1)=0</math> for <math>t &gt; T1+T2</math>.</p> | 03    | CO2 |
|        | <p>b Compute <math>u[n]*u[n]^* * u[n]</math>, the convolution of K unit step functions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 01    | CO1 |
| Q 4    | <p>a An LTI system has impulse response <math>h(t) = e^t u(-t)</math>.</p> <p>i. Is the system stable?</p> <p>ii. Is the system with impulse response <math>h(2-t)</math> causal?</p> <p>iii. Compute the unit step response of the system with impulse response <math>h(2-t)</math>.</p>                                                                                                                                                                                                                                                                    | 03    | CO2 |
|        | <p>b We define a new signal, <math>z(t)</math>, a function of two signals <math>x(t)</math> and <math>h(t)</math>, as</p> $z(t) = \int_{-\infty}^{\infty} x(-\tau + a) h(t + \tau) d\tau$ <p>Express <math>z(t)</math> in terms of <math>y(t) = x(t) * h(t)</math>.</p>                                                                                                                                                                                                                                                                                      | 01    | CO3 |
| Q 5    | <p>a Given the periodic signal</p> $x(t) = \sum_{n=-\infty, n \text{ even}}^{\infty} [u(t - 3n) - u(t - 3n - 1)].$ <p>i. Find its fundamental frequency</p> <p>ii. Find the Fourier coefficients of the exponential form</p>                                                                                                                                                                                                                                                                                                                                 | 03    | CO3 |
|        | <p>b A periodic function <math>x(t)</math> with time period T has Fourier series coefficient sequence <math>X_k</math>. If <math>x(t) = -x(t)</math> for all t, comment on the nature of <math>X_k</math></p>                                                                                                                                                                                                                                                                                                                                                | 01    | CO2 |